

Feature Engineering Summary

Milestone 2: Advanced Data Analysis and Feature Engineering

This document outlines the key feature engineering work performed in this milestone to enhance the model's predictive performance. The focus was on crafting new features, transforming existing ones, and preparing the dataset for advanced modeling.

1. New Feature Creation

1.1 Derived Metrics

- `points_per_transaction`:
Created by dividing `points_in_wallet` by `avg_transaction_value`.
Purpose: Indicates customer engagement per transaction and may reflect loyalty behavior.
- `transaction_value_per_time_unit`:
Derived from `avg_transaction_value / avg_time_spent`.
Purpose: Captures spending efficiency and engagement in relation to time spent on platform.

1.2 Time-based Features

- `last_visit_hour`: Extracted the hour from `last_visit_time`.
 - `last_visit_time_of_day`: Categorized `last_visit_hour` into 'Morning', 'Afternoon', 'Evening', and 'Night'.
Purpose: Helps understand temporal behavior of users and patterns that may correlate with churn.
 - `last_visit_AMPM`: Mapped hours to 'AM' or 'PM'.
Purpose: Simple temporal binary feature.
 - `joining_day_name`: Extracted day name from `joining_date`.
 - `is_weekend`: Flagged if the joining day was Saturday or Sunday.
Purpose: Helps analyze if weekend joiners behave differently.
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2. Categorical Feature Encoding

2.1 One-Hot Encoding

Applied to non-ordinal categorical columns except `joining_date`, `last_visit_time`, `membership_category`, and `feedback`.

Purpose: Allows categorical data to be used in most ML models.

2.2 Ordinal Encoding

- `membership_category`:

Encoded based on loyalty hierarchy:

`'No Membership' < 'Basic' < 'Silver' < 'Gold' < 'Platinum' < 'Premium'`.

Purpose: Retains ordinal relationship in numeric form.

2.3 Sentiment Encoding for Feedback

- `feedback`:

Mapped to:

- `1` : Positive feedback
- `0` : Neutral feedback
- `-1` : Negative feedback

Purpose: Quantifies sentiment to improve model interpretability.

3. Feature Transformations

3.1 Mathematical Transformations (applied to all numeric columns)

For each numeric feature, the following transformations were generated to address skewness and enable better model learning:

- Square Root (`_sqrt`)
- Square (`_square`)
- Logarithmic (`_log` using `np.log1p`)

Example: For `avg_transaction_value`, we now have:

- `avg_transaction_value_sqrt`
- `avg_transaction_value_square`
- `avg_transaction_value_log`

3.2 Selected Scaled Columns

The following features were identified for scaling and transformation:

- `age`
- `days_since_last_login`
- `avg_time_spent`
- `avg_transaction_value`
- `avg_frequency_login_days`
- `points_in_wallet`
- `points_per_transaction`
- `transaction_value_per_time_unit`
- `last_visit_hour`
- `avg_transaction_value_square` (added due to strong correlation with churn)

Purpose: Standardization ensures numerical stability during model training.

4. Feature Combination (Pipeline Ready)

A custom transformer and helper function were prepared to generate:

- **Categorical combinations:** Creating compound features from multiple categorical fields (e.g., `region+preferred_offer_types`), increasing feature richness.

Purpose: Captures interaction effects between categories which may be predictive.

5. Expected Impact on Model Performance

- **Increased predictive power** through domain-specific features (e.g., `points_per_transaction`).
- **Improved handling of skewed distributions**, enhancing model convergence and performance.
- **Sentiment encoding** provides compact representation of textual feedback.
- **Temporal patterns** help model behavioral trends tied to churn.
- **Pipeline readiness** ensures scalability and reproducibility in future model development.

